

Jason Jaeseung Kim

Biomedical Engineering Undergraduate | Medical Deep Learning and Medical Imaging

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RESEARCH FOCUS

Deep learning for medical and healthcare applications, with experience in 3D medical imaging, quantitative analysis, computational MRI, patient communication, and efficient inference. Interested in designing reliable models and experiments that turn biomedical data into clinically meaningful information.

EDUCATION

Yonsei University - B.S. Candidate in Biomedical Engineering

Mar 2021 - Expected Feb 2027

Mirae Campus | Second Major in Electrical and Electronic Engineering, Sinchon Campus

GPA	4.50 Scale	Percentile	Department Rank
4.03 / 4.30	4.18 / 4.50	97.90 / 100	1 / 49

Selected coursework: Image Processing and Applications; Digital Signal Processing; Signals and Systems; Electromagnetics I; Digital Logic Laboratory; Computer Programming; Engineering Mathematics; Deep Learning Laboratory; Public Health and Medical Big Data Analysis.

RESEARCH EXPERIENCE

Former Undergraduate Researcher, Advanced Image Processing Techniques Lab, Yonsei University

Dec 2023 - Feb 2026

Advisor: Prof. Joon Yul Choi | Mirae Campus

Deep Learning-Based Quantitative T1 Estimation | First-Author Manuscript in Preparation

- Designed experiments comparing input conditions, preprocessing and harmonization strategies, model configurations, and internal/external validation for 3D quantitative T1 estimation.
- Built automated volumetric preprocessing, model-training, baseline-reproduction, and tissue-wise evaluation pipelines.
- Contributed to drafting a multi-year government-funded research and development project proposal and supporting literature review; prepared the conference abstract and oral presentation; preparing the first-author manuscript, figures, and tables.

Quantitative MRI Study of Overwork | First-Author Manuscript in Preparation

- Designed and executed an SPM12 voxel-wise analysis with cohort curation, covariate adjustment, explicit masking, FWE correction, and atlas-based reporting.
- Automated quality-control summaries, group tables, and statistical reporting; drafted the conference abstract, figures, tables, and oral presentation.
- Preparing the first-author manuscript and coordinating revisions with the supervising professor.

Extending NAMER for Learning-Enhanced MRI Motion Estimation

Ongoing Graduation Research, 2026

Graduation Research

- NAMER uses a CNN-generated image prior to estimate motion before model-based reconstruction; current work is expanding evaluation to multiple fastMRI subjects and revising CNN training so the prior better supports downstream motion estimation.

Undergraduate Researcher, Bio-Metamaterials Lab, Yonsei University

Jul 2022 - Aug 2022

Advisor: Prof. Seung Ho Choi | Mirae Campus

- Participated in the initial installation and setup of an ellipsometer and analyzed the compositional and optical properties of biomaterials.
- Built a prototype tensile-strength measurement device and implemented a MATLAB-based user interface for measurement verification.

SELECTED RESEARCH OUTPUTS

- Oral Presentation and Outstanding Paper Award for the quantitative T1 estimation study, Korean Society of Medical and Biological Engineering Fall Conference, Nov 2025.
- Poster Presentation, NEURO2026, Jul 2026.

PROJECTS

Teacher-Difficulty-Calibrated Early-Exit Network

Deep Learning
Laboratory Project, 2026

- Developed an image-classification network combining knowledge distillation, teacher-difficulty guidance, dynamic early exits, and branch-fused deployment for efficient inference.

Project Lead, CareProofRecord - Deep-Learning-Based Medical Explanation Video Platform

Jun 2025 - Apr 2026

- Led the service concept and system architecture for patient-facing explanation videos designed to reduce information-transfer errors and improve patient understanding.
- Designed a long-context video-generation workflow intended to mitigate hallucination risk; received KRW 1,000,000 in university startup support.
- Explored potential clinical use cases with regional medical institutions and prospective users.

TECHNICAL SKILLS

Programming	Python; MATLAB; Bash; Linux
Medical Deep Learning	PyTorch; 3D deep learning; image-to-image regression; classification; knowledge distillation; early-exit inference; model explainability; quantitative evaluation
Medical Imaging and Neuroimaging	NIFTI; FSL; ANTs; SPM12; registration; masking; resampling; normalization; VBM; voxel-wise analysis
Statistics and Research Methods	Experimental design; baseline reproduction; data curation; covariate-adjusted GLM; FWE correction; ROI- and voxel-level analysis; scientific writing

SCHOLARSHIPS

- National Science and Engineering Scholarship, Korea Student Aid Foundation - full tuition (Spring 2023, Fall 2023, Spring 2026).
- Hankyungbum Scholarship, Cheongpa-Hankyungbum Scholarship Foundation, Yonsei University - full tuition (Spring 2022, Fall 2022).
- Academic Excellence Scholarship, Yonsei University (Fall 2021).

HONORS AND AWARDS

Research: Outstanding Paper Award, KOSOMBE Fall Conference, Oral Presentation Session, 2025 Academic: Academic Merit Awards, Yonsei University, 2021-2023 Programming: Excellence Award, Wise Coding Life Algorithm Competition, Yonsei University, 2022	Programming: Grand Prize, ASC Programming Contest, 2021 Entrepreneurship: Grand Prize, MEDICI+ Startup Competition, Yonsei University Wonju LINC+ Project Group, 2021 Entrepreneurship: Honorable Mention, WAVE-LAB Startup Competition, 2025; Encouragement Award, Post Corona-19 Startup Challenge, 2022
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